55 year old Nepalese woman, vision cloudy, eye hurts, head hurts, blurry vision.



- No surgery, trauma, not CL wearer
- Has nausea, vomiting, gets halos around lights (need to ask these)
- Not on any medications atm
- Has this happened before - if it has what was the treatment and did it work
- Can see mid dilated pupil, and also

  RE 50mmHg LE 14mmHg
  RE 6/120 LE 6/7.5

  Check if angles open, with gonioscopy and VH

  What determines if angle is closed on gonioscopy?
  More than 180 degree you cant see pigmented trabecular meshwork.

  DDx: possner (angles are open + more mild in terms of symptoms), uveitis (angles are open, cells are flare, miotic pupil instead of dilated).

  DIAGNOSIS = ACUTE ANGLE CLOSURE
  GOALS = Bring pressure down
  - Diamox (CI : Severe renal impairment, severe hepatic disease, sulfur allergy, hx of corneal graft)
  - One drop beta blocker e.g., timolol 0.5%
  - One drop alpha agonist e.g., apraclonidine 0.5%
  - One drop carbonic anhydrase inhibitor e.g., brinzolamide 1%

- If the eye is inflamed, one drop high-penetrance steroid prednisolone 1% with a further three drops at regular intervals in the first hour, then hourly thereafter
- NOT EFFECTIVE IN IOP > 40 AS IRIS SPHINCTER IS ISCHEMIC
- Need to do LPI, which is why we call the ophthalmologist, within 24-48 hours
    + Key studies for LPI – ZAP trial (almost 50% will develop angle closure in 5 years)
    + Eagle Study

Risk factors (AAC) = narrow angle, asian ethnicity, females, hyperopic

Pupil block = key mechanism for angle closure, aqueous humour (PC - AC) , iris is forwards which closes the angle. LPI is just alternate route for the AGH to go

stops flowing from
Iris plateau mechanism= iris is flat, anatomical pushed forward by CB, so angle may remain closed even after LPI, consider iridoplasty or lens extraction

- Iridioplasty = burn laser around iris

- Not CL wearer
- Nausea and vomiting
- Haloes

- Previous incidences

RE: 50 mmHg, LE: 14 mmHg
- IOP over 40, pilocarpine will not work as the sphincter is ischemic
- Predforte as inflamed
- Diamox: Call GP, n
- Bilater LPI within 2 days
    - ZAP study

Corneal oedema

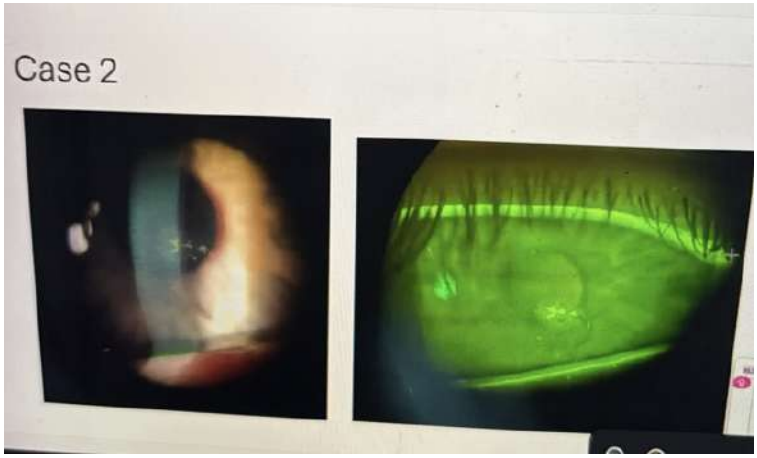
DDx:
AAC
Uveitis



What are some clinical signs indicating prior angle closure attack?

- Top left = iris atrophy (happens in mid periphery if there is high ischemia from previous high IOPs)
- Bottom left = glaucoma flexcken lmaooo (subcapsular lens opacities caused by ischemic necrosis of the lens epitehlikln)
- Top right = dilated iris vessels
- Bottom right = closed angle, can also check for pigment blocking ™, also neovascularisation, synechiae (post iris attaches to lens, ant iris to trab)

Dilation can also cause AAC.



“Feels like he had metal shavings fly into his eye”

- When did it start– 2 days ago
- Any pain or **photophobia** - sore, little painful
- **Any itch?**
- Has anything helped - Chlorsig
- No HAs, slight blurry vision
- Was wearing protective gear

- Has this happened before - first time
- No **blunt trauma** that he knows of - thinks something flew in his eye
- Has it become worse - yes, RE
- Any **discharge** - slight blur, no discharge
- No previous sickness (cold sore in his mouth week before).
- No **CL use**

The eye is sore, slightly photophobic

Next = Pupils, VA (RE 6/9 LE 6/6), motilities, pressures (RE 15 LE 16), white light and fluorescein

Photos
- Lesion is positively stained, branch like appearance

Diagnosis = Herpes Simplex (with terminal bulbs), cranial nerve 5 but in ganglion..
Zoster does not stain on fluro, no terminal bulbs (pseudo dendrites), stains well with rose bengal

Treatment = Oral Acyclovir 400mg X2 , topical aciclovir + lubricant
Switch to oral if having severe toxic reaction, stromal involvement (stromal haze)

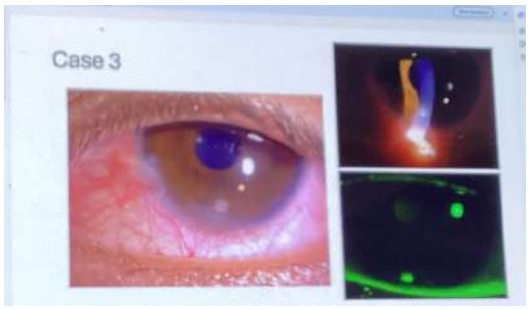
DO NOT EXPECT TO SEE INFILTRATES , NO STEROID AS SUPPRESSES IMMUNE RESPONSE (will become geographic ulcer that becomes ulceration)

If stromal or endothelial involvement use pred forte 1% X4 a day, also use oral anti-viral
Side Effects = corneal toxicity, hypersensitivity reaction, SPK

- Know the HEDS study (why u use oral antivirals as prophylaxis), use oral antiviral for oral immunocompression and elderly px cant instil drop themselves, will help resolve actual keratitis itself
- NEVER PUT A STEROID ON AN OPEN HERPES DEFECT (FAILER!)
- WILL NEED TO SAY U HAVE TO DILATE (FAILER!)
- ARN and PRN what lmnaooo at the back of the ey???

24F , vision slightly blurry in past few months esp at distance.

- CL wearer - haven't showered or slept with CLs, what type of CLs
- LE has been sore, watery and also light sensitive
- General health clear
- LE
- No itch no discharge
- Any previous sickness - no sickness
- Tried lubricants didn't help
- Hasn't happened before
- Stopped cls wear- gotten worse
- VA: RE 6/12 LE 6/7.5
- 16 12 IOPs
- White light + NAFL



- Corneal oedema
- Injected BVs, crossing limbus (pannus)

NOTE: Pannus indicates

DDx: CLPU, MK, marginal keratitis, acanthamoeba keratitis, fungal keratitis
Grade 1 cells in anterior chamber , PUK *BUT SOOOO PAINFUL

- Marginal keratitis - usually inferior
- MK

TREATMENT

- NOT chloramphenicol
- Ciprofloxacin 0.3%, loading dose every 15ms for 2 hours, then half an hour for another 2 hours, then hourly until lesion goes away.
- Corneal precipitates bad because u can confuse with true infiltration

- Ofloxacin can lead to corneal perforation (not to use in patients with elderly and those with thin corneas and those with endothelial dysfunction)
- NO STEROID (ONLY ONCE EPITHELIUM HEALS to help medicate AC reaction + prevent scarring? BUT IT SHOULD RESOLVE ANYWAYS)

Can also use dual fortified antibiotics used if more severe MK (larger infiltrate, greater size of lesion, if centrally located, also elderly people as corneal regeneration slower and more compromised compared to if they were 20 years old)

Every 30 mins separately
Cephalosporin inactivates gentamicin if used together

Beta lactam breaks cell wall of
Lmao gang

No taper for antibiotics because of antibiotic resistance- minimum 4 times a day

SCUT trial- steroids 48 hrs after ciprofloxacin or when the wound re-epithelialises to reduce scar formation

62M, routine exam, no CC, Father glaucoma, mild myopia, high BP (taking BB)

- Any trauma
- No migraine, sleep aponea, raynauds phenomenon

VA 6/6 OU, pupils PERRL (looking for an RAPD), Golman IOP 26OD 24OS, CCT (528 and 531), confrontation field , Open angles, DFE

- Anterior exam : specifically check angles,
- DFE : Optic nerve (compare size of nerves, CDR (proxy of NRR health), size of disc itself)

CDR ratio as that will give me good indication of disc margin health. I will take into account disc size as well

Note: Check CCT for IOP udner/over estimation, risk factor

Photos:

- CDR 0.8
- Inf temporal rim loss

FASTER 4 step , more likelyto drop out false negative (more reliable because of focusing issues)
FAST 2 step

- Could be an optic neuropathy, not sure which at this point,
- Compressive neuropathy, naion, aion

DDx:

- Could be POAG
- Could be OHT
- Could be secondary (uveitic, traumatic)

With reduced light sensitivity so in very deep visual field defects, would suspect that RAPD would not be a good test as his eye has reduced light sensitivity

Something stopping the light from going in??

What does a pupil test do?

-

No pallor, no RAPD,

POAG , advanced as centre involving VF.

6M – do a 10-2, see how much it is affecting central vision.

What is glaucoma?
Glaucoma is a progressive optic neuropathy , with structural damage to ONH, loss of ganglion cells and axons with field loss, associated with increased pressures

X3 theories of glaucoma

1. Mechanical
2. Ischaemic
3. Mitochondrial

Adrian phuong - westmead eye manual

Pregnant woman SLT THAN timolol
Brimonide has CNS depression risks in those below 20Kgs, could kill child?
Dorzolamide B3 pregnancy but it hasn't been tested in humans
Choose Beta Blocker (ONLY TIMOLOL) if patient is pregnant Pregnancy C

NEXT CASE

- Keratic precipitates on cornea
- Pigmented cells in focal arrangement
- PDS (thick texta line)
- Loss of pigmented pupillary ryff

10-40% of PDS patients

PXF, scandinavian, 70%, 30% of scandinavian women have this issue

Dont see much of this in australia

PBS REESTRICTED - NEED TO BE SEEN BY OPHTHAL WITHIN 3 MONTHS OF COMMENCEMENT

COMBINATION RESTRICTED FOR monotherapy

12M scripts that px for going overseas can give, has to be stable but shoiu;d be seeing px in 6M.

- Pregnancy
- Contraindication vs precaution
- PBS restrictions
- Adherence
- All glaucomas are bilateral just different stages (physiologoical NOT TRAUMATIC), all hereditary or genetics

- History be efficient and decisive abt what ur asking
- Which eye, what position
- Shape, depth and size
- Staining use pupil size
- Medications know dosages, concentration

Antibiotics don go under X4 a day, not , acuate angle closure no

Angela and Calvin Cases

CASE 1

What do you want to know?

F: need to know recurrence
O: 2 days ago
L
D
A: nausea and vomiting, lights look like halos
R: have you tried anything to help?
Q



OD VA: 6/120 OS: 6/7.5

Slit Lamp Exam
Lids/Lashes
Cornea: Oedema
Conj: Hyp 360

ACUTE ANGLE CLOSURE

DDx:

- Anterior Uveitis

TREATMENT

- DIAMOX
- LOWERING DORPS
- CONSIDER PRED FORT
- IOP > 40 → PILO WONT WORK SINCE Sphincter muscle is ischaemic

Risk factors for needing LPI

- Narrow angles → gonio → (Check for pigmentation, neovasc)
- ZAP STUDY
- EAGLE study: lens extraction
    - Presbyopic
    - Cataracts
- COMPARE LPI AND LENS EXTRACTION
- Iridoplasty: burn iris and physically widen angle
    - When consider Iridoplasty over iridotomy: Plateau iris (Go for iridoplasty or lens exchange over iridotomy)
    - LPI will not fix the anatomical crowding
- DO NOT DO LPI

Feature	Laser Peripheral Iridotomy (LPI)	Laser Peripheral Iridoplasty (LPIp / ALPI)
Primary Mechanism	Creates a small full-thickness hole in the iris to equalize pressure between the posterior and anterior chambers.	Uses low-power, long-duration contraction burns in the far peripheral iris to shrink the stroma and pull the iris away from the drainage angle.
Target Condition	Pupillary block (where iris-lens contact restricts aqueous flow).	Non-pupillary block mechanisms, such as plateau iris syndrome, anterior ciliary process rotation, or persistent narrowing post-iridotomy.
Laser Settings	High-energy, short-duration/photodisruptive pulse (e.g., Nd:YAG or Argon).	Low-power (approx. 200 mW), large spot size (150–200 µm), long duration (0.5 s) Argon laser burns.
Corneal Visibility Need	Requires a relatively clear cornea; more difficult to perform if severe corneal edema is present.	Can be performed even when corneal clouding or severe acute swelling is present, often acting as a temporary bridge.

PTM

Top left: Iris atrophy

Bottom right Signs of previous closure attack: Glaucomflecken

Top right : dilated iris vessels

Bottom right: Closed angles



Other signs of closed angle attack

- PAS : peripheral anterior synechiae

Mechanism of angle closure

- Pupillary block
    - Drops, drugs (antipsychosis)

    - Uveitis: synechiae
    - Secondary causes: NV, pigment clogging up TM(?), Drops
-

PHOTOS
TOP LHS: IRIS ATROPHY
TOP RHS

Bottom lhs: Glaucomflecken

Case 2

32 y/o male red watery eye → FB sensation, vision is slightly blurry
Occupation: Metal Grinder

GUYS ASK IN FOLDARQ FORMAT (Frequency, onset, laterality, duration, associated symptoms, relief, quality)

- Onset 2 days, works metal grinder
- No discharge, itch, nausea, haloes, Little sensitivity to light
- Chemist prescribed chlorsig → did not help
- Unilateral redness
- Associating Symptoms: irritated not really painful
- Has gotten a bit worse
- Redness / irritation → Constant
- Not CL wearer, was wearing safety goggles
- First time happened

For Red Eye Ask

- Discharge, Pain, Itchiness, Nausea, Photophobia, ASK VEGETATIVE HISTORY

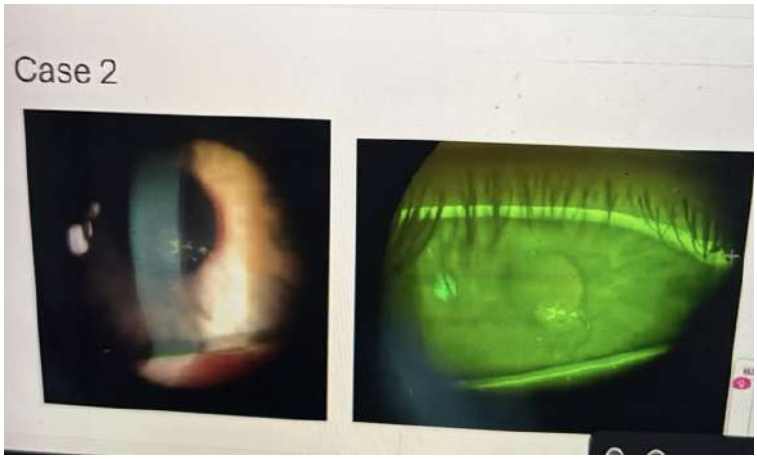
Tests

- VA OD 6/9 OS 6/6
- Motilities SAFE, Pupils PERRL
- IOP OD: 15 OS: 15
- Sit Lamp Check
    - Cornea:
        - NaFl
        - Irregular border epithelial defect → Dendrites/Pseudodendrites
        - Location: Inferonasal but pretty central
        - SATELLITE LESION around main dendritic ulcer
        - Stromal haze → oedema

- Active Oedema = Active inflammation (expected with microbial infections)

Diagnosis/DDX

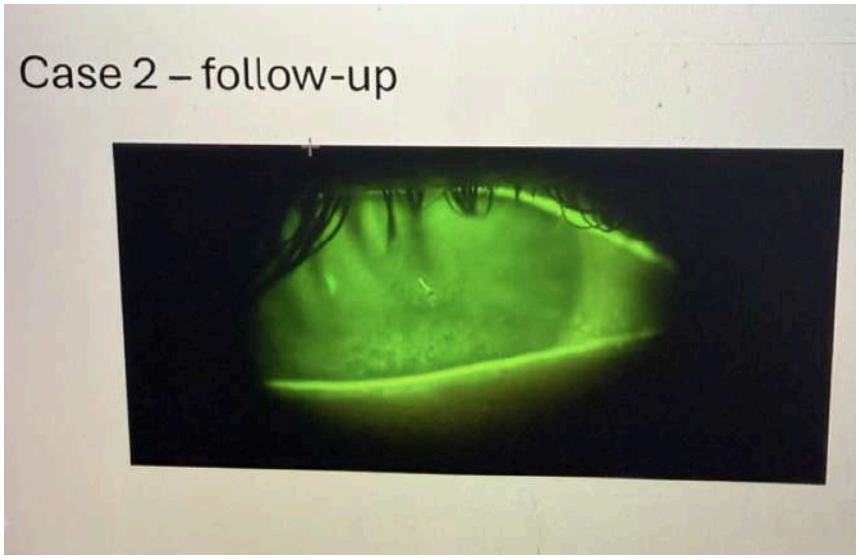
- HZO
- FB
- HSK vs HZO (more skin lesions)
- Pseudodendrites and dendrites → Pseudod
    - HZO is prodrmal → follows along midline and unilateral. Skin lesion over bod -
    - Trigeminal nerve along ophthalmic division: scalp division 1
    - HSV stains strongly on NaFl HZV doesn't. HZV stains strongly on rose bengal
- ACANTHAMOEBA KERA - signs disproportionate to symptoms (symp pain sign mild)
    - , free flowing water like ponds
- FUNGAL KERA - satellite lesion is very fungal characteristic - no feathery edges, no ring infiltrate, NEED TO SPECIFICALLY ASK ABOUT CONTACT WITH VEGETATIVE MANNER
    - Symptoms disproportionate to signs for fungal and esp acanthamoeba
- BACTERIAL KERA



Management

- Topical aciclovir 3% ointment 5x/day
    - Adverse reaction: toxic to ocular surface, would you stop the treatment? No as infection won't get better
- What case would you use orals for herpes simplex? (need approval with GP) Stromal yes, uveitis no
    - If non compliant with eye drops
    - Elderly (dexterity issues – difficulty with using drops)
    - Toxic effect
    - Immunocompromised
- Aciclovir 400mg 5x/day for zoster → bid for simplex
- Vavicyclovir 500mg
- Famocyclivir (might not be available in australia)

- HSK BID for orals



Prognosis (how long to heal)

- 7 days/ up to 10day mark
- Patchy green/black pattern
- Black areas = negative staining. Not true staining because no GLOW. slight oedema /corneal irregularity
- More concentration inferior: due to ointment pooling down (from gravity? Is hat what hes sayig?
-

HED STUDY - prophylaxis - when its indicated

HEDS1:

- Prophylactic oral 38% reduction for HSV
- Topical steroids may speed resolution of HSV stromal, but do not affect the VA.
- Systemic aciclovir was not useful for HSV stromal already receiving topical steroids.

HEDS2:

- Prophylactic oral aciclovir had a 41% reduction in any herpetic episodes and a 50% reduction in stromal keratitis
- Long term oral aciclovir is well-tolerated.

Environmental stressors, sunlight, and CL use has not been shown to be associated with recurrence.

Get rid of epithlium
ted

Ask about cold sores (zoster)

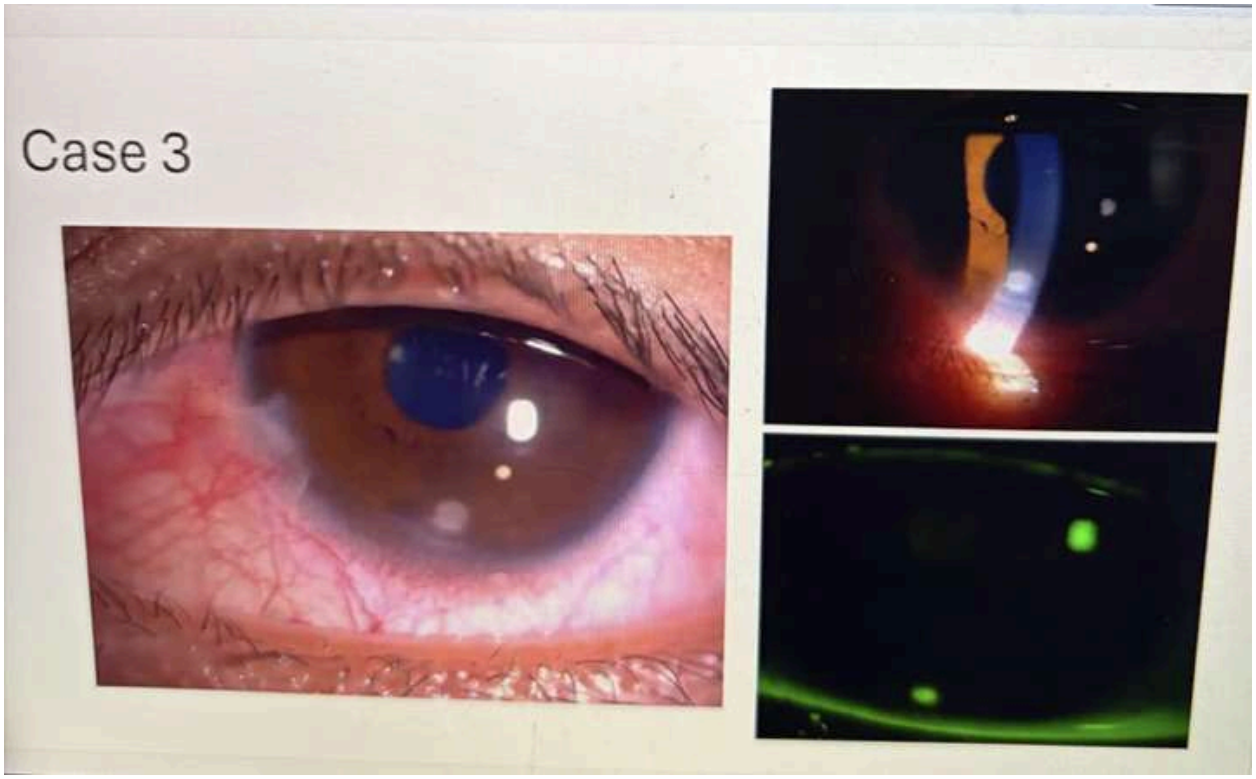
Oh my god is this HSV Stromal Keratitis with ulceration…→ oh nvm lol…… sorry

- Would be a lot more oedematous

Case 3
24yo Female came present with vision blurry??

- Onset: gradual onset, probably worsened 2 days ago
- LE sore, watery, light sensitive
- Notice blur with both eyes open
- Yes CL wearer – dailies
- Gen health clear, oc health clear
- No discharge
- No recent sickness
- No allergies, no itch
- Consistent blur, does not get worse throughout day

VAs with old glasses: R6/12 L6/7.5
PH R6/7.5 R NIPH
IOP R16, L12



Slit lamp
OS

- Pterygium nasal
- Limbal redness
-

- Infiltrate inferior + Pannus
    - NaFl: Inferior infiltrate base stains → Ulcer with infiltration
        - “With infiltration” because u can see the whiteness on white light

DDX
- Marginal keratitis i
- CLPU (more inflammatory, shouldnt have AC reaction)
- MK
- Peripheral ulcerative keratitis (autoimmune)

Fluoro mono vs dual therapy (mono if peripheral, dual if central)
Cipro vs oflox
Precipitation in cipro can mask lesions (telsl you px is using hte drops)
- Every 15 min for 6 hours, hourly for 6-24 hours
- Decrease dose after re-epitheliased
- STEROID: SCUT trial no short term improvement but long term improve scar
    - 48 hour after cipro
- Discuss aminoglycoside class, Dual therapy
-

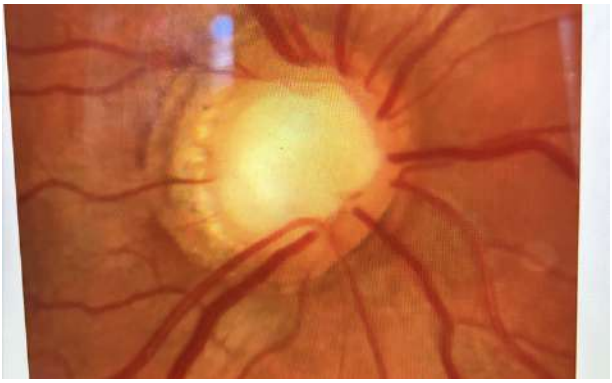
Steroid if:
1. Central involving or for px comfort but dose correctly (qid)
2. Central → predforte
3. Peripheral → no need steroid but use fml over pred forte

Blake Cases

Case 1: no CC; routine check (-) concerns
62 yo
No asthma, no steroid use
Hypertension (well managed now), beta blocker
Dad has glaucoma

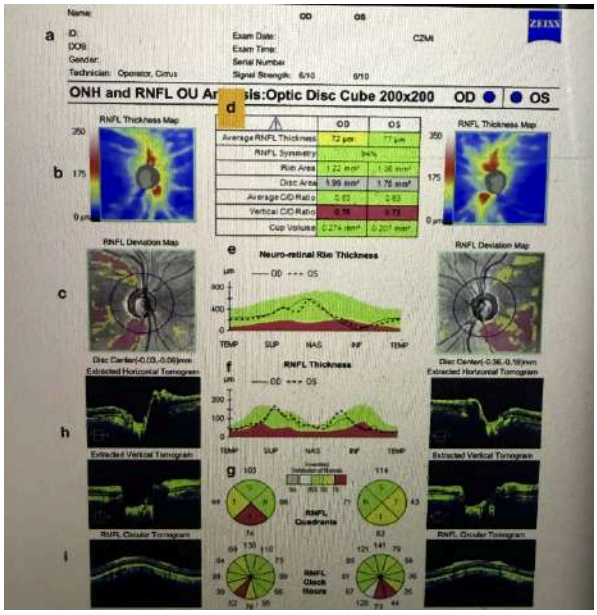
- LEE: no one mentioned anything just here for glasses really
- No history of trauma =
-

IOP R26, L24
CCT: 528, 531
Angles: Gonio grade 3 or 4
Disc: 0.8, notching (inferior temporal), PPA, lamina pores



OCT: inferior temporal RNFL wedge defect

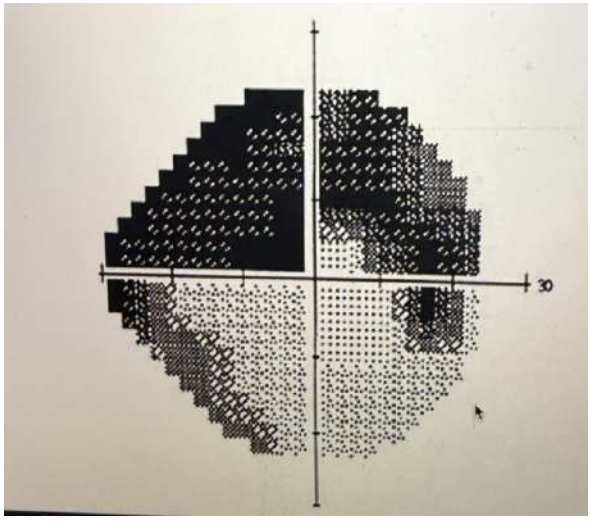
- Don't jump straight to problem, just describe what you see first
- Eg. rnfl thickness map, heat map looks like potential inferotemporal area thinning corresponding to notching, TISNT, seems different in inferior between 2 eyes, tomogram flagging inferior region
- Structure agreement across scans



24-2 sita faster

- First time, baseline
- Potentially superior defect
- Why faster, no fast or other
    - Reliability of all tests, some less reliable as they don't check
    - Front loading fields research

- Reliability indices: FP, blind spot monitoring, no FN?, fixations losses
- MD, PSD
- Raw numbers (sensitivity points), heat map/probability map away from age norm, p value represented in colour, mean deviation plot (used to calculate MD), then pattern standard deviation (separate any gross defect from localised defect, PSD calculated off this)
- GHT, 5 points superior and inferior, any difference = not wnl



- Superior arcuate defect(quite wide and deep) Indistinguishable, superior nasal quadranopia (concerning), inferior nasal step extending into possible arcuate
- Can't say its glaucomatous straight away
- Structure function concordance

DDX (optic neuropathy of some kind)

- POAG
- NAION
- Vascular occlusions (possibly before HTN was well managed)
- Ocular hypertension
- 2ndary: PXF, PDS uveitis, traumatic
- Closed angle glaucoma
- Compressive, ischemic, inherited
- Signs for non glaucomatous optic neuropathy: RAPD? Would px have rapd if it was glaucoma? Probably due to reduction in VF
- Non glaucomatous: DISC PALLOR, indicative of something BAD

10-2 as it is quite a central defect

Dx: POAG

- Late stage, MD worse than 12

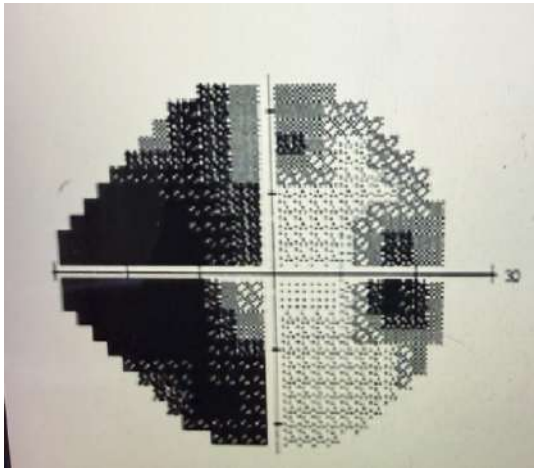
Management

- Prostaglandin analogue increases uveoscleral outflow
- MOA: remodelling the extracellular matrix
- Side effects: darker + longer lashes, redness, periocular fat atrophy/darkening of skin, iris darkening, stinging, herpetic reactivation?, dry eyes
- Latanoprost 0.005
- 5 repeats. 6 months, not 12 months (needs to be stable)
    - Monitor for progression
- PBS requirement for latanoprost: need to be diagnosed by an opthal (within 4 months of diagnosis)
-

Follow up

- 6 week IOP, check for 25-35% reduction. Aiming for 30%, so 18
- Timolol 0.5% decrease aqueous production… however currently on a beta blocker. Punctual occlude ,
- Brinzolamide 1.0% carbonic anhydrase inhibitor
    - Contraindications: sulfur allergies, renal complications
    - Pregnancy? SLT
- SLT vs Drops
    - SLT, repeat 3-5 years, just as effective as drops

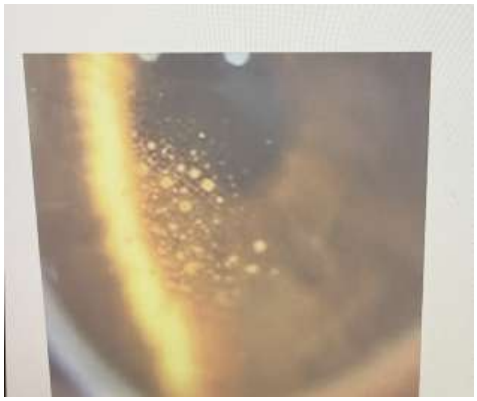
- COMPLIANCE issue with drops. SLT

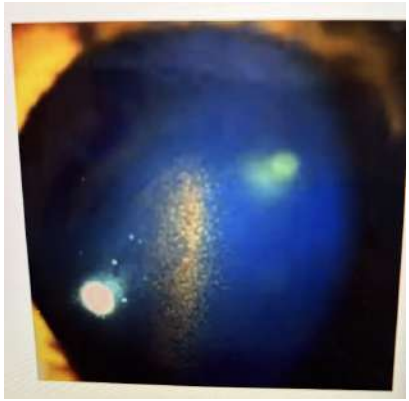


Px returns for 6wk F/U, complains of blurry vision and this is the VF.

- Compressive lesion on optic chiasm?

New Px:
KP



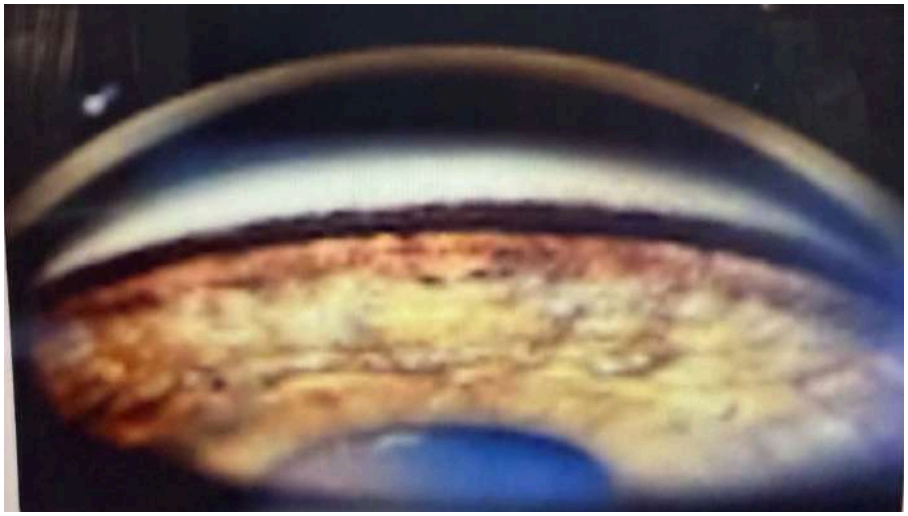


Krukenberg spindle: could be sign of PDS

- Gonio to check trabecular meshwork pigment?
- acute/ recent inferior angle pigment. IOP increased
- Chronic more superior angle
- The iris bows

Check for midperipheral TIDs
Epidemiology:

- Young male
- Myopia
- Fam Hx
- Vigorous exercise, IOP spike following



Gonio: either PDS or angle recession

- Iris concavity
- Heavy pigmentation inferiorly

Prognosis

- 10-40% risk of glaucoma conversion
- Burn out effect as after a while there is no pigment left

3 types of glaucoma: 3 theories

1. Mechanical (IOP)
2. Vascular (NTG?)
3. Cellular molecular (mitochondrial)

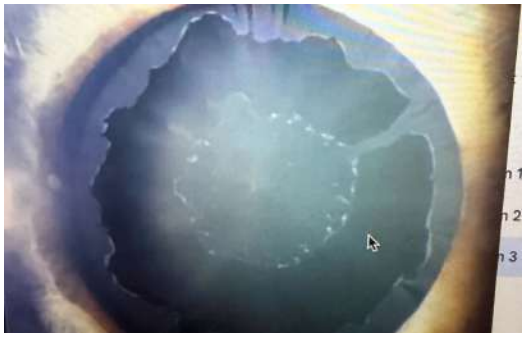
Best treatment for PDS:

- SLT vs drops: SLT less effective - pigment in angle absorbs laser; often will need repeat laser
- LPI: to reduce iridozonular contact. Won't fix glaucoma, will stop further pigment liberation but there is already pigment there
- Trab more likely in these px

PXF



- Loss of pupillary ruff
- TID
- White pseudo exfoliative material on anterior surface of lens



- Bulls eye pattern: Worse than PDS
- 60% risk of progression to glaucoma (50 is POAG)
- Scandinavia, old women 70+, uv exposure, cold climate, high latitude. Genetic predisposition but if they move to Australia its better. Less common here
- Rate in Scandinavia: 30% in over 70s
- Management: Same drops as before, very careful, greater diurnal fluctuations, broken up schawbes line in gonio, does not respond very well to medication
    - Combo therapy more likely later on

Things to revise

- STUDIES: **adrian fong westmead eye manual**, chapter showing every study for every condition, OHTS, main outcomes
- Pregnancy
    - Beta blockers neonatal brady cardia
    - CAI
    - PGA → contractions
    - Brimonodine → neonatal depression?
    - Aus pregnancy classes (B3 (no study) is worse than C. C formally tested and B3 no study)
- PBS requirements
    - Single - dx confirmed by ophthal within 3M

    - Combo - not stable
- Adherence

Tips
Image: talk about what you see. Can correct yourself.

2 major cases didn't go through today
- Hyphema treatment, intense
- Optic neuritis: oral prednisolone (isnt it like methylprednisone too or smth), ONTT trial

---

CASE 2A
21 yo male irritated itchy eyes
F: first time
O: 6 days ago, worse in the morning, always there
L: Right then left, now bilateral
D:
A: little pain and light sensitivity, eye rubbing
R:
Q:
(-) CL wearer
No recent illness or allergies, recently moved to australia from pakistan with with family





Pupils motilities ok
IOP R 22, L 21
VA 6/7.5, 6/6. Mild astigmatism R>L

- Lid oedema and erythema
- Inferior papillae(red blood vessels)/follicles (white head, white blood cells).
- No discharge, (+) watery eyes
- Fluorescein, cornea clear
- AC quiet
- Upper eyelid eversion: similar appearance

DDx:
- Seasonal/Perennial allergic
- Dry eyes

- Viral conjunctivitis
- Preseptal cellulitis. Orbital cellulitis (no pain on motilities, ruled out). Usually discharge for both
    - Hx of hordeolum (no hx)
- Contact dermatitis

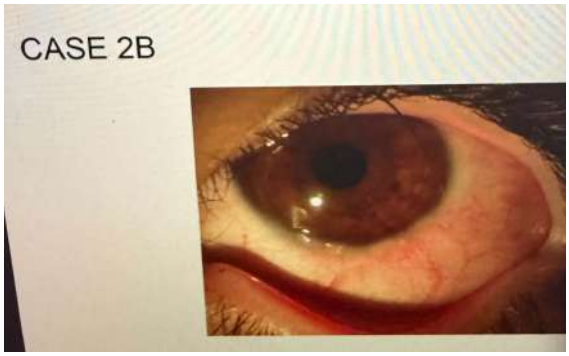
Mx

- Zaditen preserved ok since pf no available rn 2x daily (narrow therapeutic range. Too much = reversal and histamine release)
- Ocular lubricants (chilled?)
- Cold compresses
- Avoid allergen (lifestyle moditifcation) such as showering before bedtime, freq wash clothes/pillowcase, avoid likely allergens" dust, pollen, animal fur
- Mast cell stabiliser before season starts (long term management)
- Follow up no improvement
    - Check compliance, recheck diagnosis
    - NSAID: used in conjunction with the zaditen, however stings
    - Steroid? FML 0.1% 4 x daily for 1 week and review
        - Ideally low frequency and low potency
        - Maybe 2x daily instead?? As 4 is not considered low frequency?
    - Oral antihistamines
    - Ciclosporin later?
    - Zaditen will take 2-4 weeks to work

Case 2
11yo boy
No health/med history/ocular infection/surgery
Bilateral itchy eyes onset 1 week, watery discharge
(-) CL wear
(-) recent sickness
(+) discomfort

VA: OD 6/9, OS 6/7.5 = mild astigmatic rx
Motilies: safe OU
Pupil: normal
IOP: OD 16 OS 14

Clinical findings:







- Raised gelatinous thickening. Go from out to in
- Conjunctival hyperaemia temporally and inferiorly

- Inferior palpebral conj: oedematous and hyperaemia
- Velvety → soft and slightly oedematous appearance
- White mucous strand: white and ropey (not pseudomembrane)

- Large papillae: cobblestone appearance
-

DDX

- VKC: horners tranters dots, young male
- AKC
- Viral conjunctivitis
- Giant papillary conjunctivitis
- Marginal keratitis: due to limbal dots, however, no blepharitis, usually located in peripheral cornea, however these dots on the limbus
- SLK

VKC

- Type 1 hypersensitivity reaction, IgE
- Young males in warmer conditions DRY conditions
- Atopic or inflammatory? Atopy fam hx seen commonly
- Intense itching, watery, photophobic (due to cornea aspect)

- Grade subtype: shield ulcer??
    - Moderate to severe
-

Management
- Combo drug. Zaditen 2x daily
- Steroid: high dose pulse corticosteroid. Don't need deep penetration
    - FML (safe as more likely to be steroid responders in children), worried about growth, cataract, glaucoma
    - 4 x day
    - Roughly 2 weeks
    - Iop spike more likely in 3-6weeks
- GP systemic
- Refer for allergy testing

Prognosis
- Puberty will help outgrow

Case 4
25 yo male
F: 3 days increasing red, achy, sore (7/10), light sensitive OS. First onset
VA slightly blur
Mild tearing (not really discharge)
Walked in with sunnies (photosensitive)
OTC drops haven't helped (chlorsig eye drops lol)
(-) CL wear
(-) trauma with vegetative matter
(-) recent sickness
(-) nausea
(-) systemic involvement - no skin lesions/rash, no autoimmune conditions
(+) occasional sore back

BCVA: OD 6/6, OS 6/12 NIPH
Pupil: OD normal, OS irregular but still
Motilies: no pain on movement
IOP: OD 16, OS 11



Conj: diffuse hyperaemia, nasal side more red
Pupi: irregular pupillary margin nasal temporal
Hypopyon
Circumlimbal flush/ciliary flush
Hazy cornea



KP; on endothelium
* try start broad lol, dont be too specific: describe as deposits, lesions (v broad)
    - Central to inferior location
Granulomatous vs non

Additional test:
- AC assessment (hypopyon, g4 cells, g3 flares)
- Check angles and pressures prior to dilation
- Dilate to rule out posterior involvement
- Fluorescein (no staining)

DDx:
- Unilateral acute anterior uveitis
- Endophthalmitis (rule out if posterior is clear) no surgeries or penetrating eye injuries (more likely to be iatrogenic or traumatic
- Masquerade syndrome
- Acute angle closure (angles were open, iris would be mid dilated (uveitis is mid dilated)
- Bacterial keratitis (no ulcer, no cl wear)

MX

- Corticosteroid: predforte, high penetration and high potenety, has phenylephrine, help to resolve synechiae. Once every 1-2 hours until sun grade drops 2 or to 0?
    - Loading dose as hypopyon, every 15 min for first hour a,d every 30 min for 2 hours
- Atropine 1.0% paralysis iris schincter muscle so there is less pain and phophobia as pupil dilates. Stabilise blood aqueous barrier and break posterior synechiae
- Phyenylephrine 2.5% once max effect of cyclo has not helped
- When referral to ophthalmology?: quite severe as hypopyon
    - Bilateral
    - Not responding to treatment
    - Underlying systemic cause

5 day review

- Cells g2, flare g1, hypopyon resolved
- Reduce steroid: 2 hourly
- Remove atropine (atropine lasts for 1 week)

In another week

- Cells 0.5, no flares, no hypopyon, no flares
- VA: 6/6 OU
- IOP: 29 : beta blocker or CAI, no pGA as pro inflammatory
    - Bradycardia and asthma no beta blocker

6 months later similar episode in other eye

- Start treatment
- Referral for systemic workup: C protein coin, non inflammatory mediators, ESR, HLAb 27
- Additional imaging: chest X-ray (TB - lesions in the lung)
-

Sukanya iyer

- Big broad statements
- Can email her unsw email if any questions - is she a supervisor??? no idea
- Explain thought process
- HX: efficient and decisive. Dont go round in circles, go to the

Which eye, what position
Lesion: shape depth, size
Staining: use pupil size to estimate

Dosage: be clear with schedule (MK? Wud Day 1, day 2) + concentration of drugs

Red flogs

- Antibiotic dont go under minimum inhibitory no less than X4 a day
- No gonio with acute anterior closure , u dont have rime for that oki
- Miss contridnicaiton
- Prostglandind when inflammatory
- Its a red flag if you DONT dilate with herpes
- Oral meds?
- MK: dual fortified (and why)
- MOA, why those over the initial therapy, justify everything you say
- Set line, standard questions for history (especially for red eye)
- Always ask VA (VA FIRST) , pupil, motilities, IOP

Red eye list of qs:
Pain, photophobia, discharge, itchy, CL use, hx of trauma, it is blurry

- History be efficient and decisive abt what ur asking
- Quick history for red eye
- Which eye, what position
- Shape, depth and size
- Staining use pupil size for reference
- Medications know dosages, concentration, know MOA
- Know how to justify why it is A and not B
- Start with a broad sweeping statement
- Can ask is this all he info im getting

.

“White ropey discharge” for VKC

CASE 2B

- For papillae = GPC (need to ask about CL use even if young), SLK, AKC, VKC

CASE DIAGNOSIS = VKC

Steroid = Fast taper (3,2,1)

- Risk of recurrence = (child is 11), so we do think it will come back until he reaches puberty. Long term try allergen avoidance, lid scrubs, cold compression

CASE 4

25M increasingly red, sore LE that is light sensitive. Vision is a little blurry, mild tearing and OTC eye drops have not helped.

H/S

- Started 3 days ago
- Started using chlorsig
- Not a CL wearer
- Hasn’t had this before
- Not painful, just sore
- No allergies
- Not taking any medications
- Could do a review of systems (joints, rashes, fever, night sweats).

PHOTOS (always work from out to in)

- Lashes clear
- Temporal conjunctiva quieter compared to nasal
- Diffusely hyperemaic
- Pupil is irregular shape (most likely from synechiae)
- Mild hypopyn (also seen in endophthalmitis)
- KP (large white lesions or depoists on the endothelium)

VA 6/12

OS 11mmHg OD 16mmHg

DDx:

- Uveitis (acute anterior uveitis) – need a dilation

- Episcleritis
- Scleritis
- MK
- Endoopthalmitis (VERY PAINFUL and px is reporting this as mildly sore)
- Acute angle closure (know that it is not because pressures are good, no corneal oedema, not in severe pain)

Causes for acute anterior uveitis = Idiopathic, auto-immune disease (HLA-B27), infective (sphiliis or TB)

Management = Steroid of predniosIone actate 1% (pred-forte), 1 drop hourly until 2 step improvement on sun grading scale ,
Loading dose every 15 mins for the first hour and then every 30 mins for 2.
Predforte is suspension so need to shake
Use cycloplegic for pain + break synechiae . Atropine has longer half life and interms of efficacy.

Review = 1 day review. Taper when we see 2 step improvement on sun grading scale. Review every 5 days for the taper

If clinical signs come back = restart regimen + loading dose + increase taper

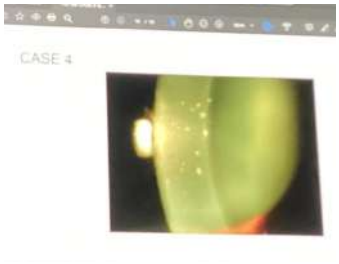
Been reviewing patient every 5 days and gotten to 3 week mark, but IOP now 29 in OD

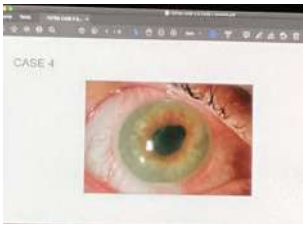
- Patient likely a steroid responder
- Uveitis has now completely resolved
- Start tapering
- Start beta blocker + briominidine (COMBIGAN) - no PGA is pro-inflammatory . Combo drugs just to get the pressure down but can also use timolol by itself.

REFER IF BILATERAL + SEVERE + POSTERIOR INVOLVEMENT + NOT RESPONDING TO STEROID + IF HAD ANY RECENT SURGERY

Say come back 6 months later but in other eye

- Refer for inflammatory marker testing , look for blood, chest x-ray, HLA-B27 immunomarker test, IGG for syphilis


CASE 4


CASE 4

Breakout room 3

Case 1 POAG

- 62 male father myopia, routine eye exam
- Dad glaucoma: When was dad diagnosed?: Unknown
- Hypertension (Betablockers) asthma
- No steroid use
- Migraines, reynauds phenomenon
- VA? 6/6 both eyes
- Motilities, pupil
- Slit lamp, gonio (open angles)
- Pressures GAT (R 26 L 24)
- CTT R 528 L 531
- DFE → disc, notching, vertical CDR, RNFL drop out, drance haem, deep cupping
    - Look for disc colour

Right eye

- 0.85 CDR
- Inferior temporal RNFL loss and drop out
- Sit back and squint to see notching??

VF

- no FN for 24-2 Fast?? Double check

Ddx:

- Superior nasal quadrantopia
- Superior arcuate

Even pigmentation
Nice and open
No focal posing
No PAS

Ddx
Neurological
POAG → staging: advanced

- Progressive optic neuropathy with structural nerve head damage and corresponding VF loss. With increased IOP sometimes
- Multifocal causes
    - Increase IOP -> barotropmic damage
    - Hypereperfusion
    - Metabolic dsynfction
- ??mitocondiral??
- May ask about:

    - Mehcanial theory:
    - Vascular theory:
    - Cellulor or molecular theory:

Pros and cons of medication

- Compliance
- Side effects
- Hard with poor dexterity
- PGA -> increases uveoscleral outflow
    - SE: orbital fat loss, lash thickening and lenghtling, iris pigmentation, pro-inflammatory (avoid in uveitis, Hx herpes, CMO)
- No BB (already on) timolol 0.5%
    - Exact script you would give
    - PBS
    - Can do repeat 2 but
    - To meet PBS restruction requirements → need opthahl appointment to
- If IOP doesnt fall
    - Check compliacne
    - Check technique
    - Consider surgery or adding another drop
- How does SLT work and when do u choose
- How often review and when measure
- What is the evidence for treating glauc (landmark trials)

Returns for IOP check 6-8 weeks after

- Field shows vertical midline respectingd defect
- Risk of POAG and progression → univariant: OHT study: Large CDR, thin CCT, initial high VF loss (high MD), FHx (first degree relative - siblings > parents > grandparents), baseline IOP
    - Multivarent: Diabetes, african ethnicity, myopia

Case 2

Photo: KP, krukenburg spindle

- PDS
    - Mechanical bowing of the iris which rubs against lens zonules
    - TID seen
    - Pigment on gonio
    - high/ fluctuating IOP after exercise
    - Zennaey line??????
- Epidemiology: young myopic (larger iris) males, bilateral but asymmetric presentation?
- *Which study?* How many cases progress into glaucoma? 20-50%
- burnout phase → pigment reversal sign (pigmentation of superior angle > inferior angle)
- Treatment:
    - SLT is less durable (due to heavy pigment)
    - LPI to reduce iridozonular contact and further liberate pigment
    - Filtration surgery?
    - PGA increases uveal outflow by remodelling extracellular matrix regulation in the ciliary muscle, relaxing the muscle and improving fluid outflow.
- Not much evidence around PDS. A lot of the Mx is anecdotyal. High quality RCT evidence of PDS is limited
    - 10 year study LPI may lower
    - Variable long term IOP lowering

Case 3

- ECM at pupillary margin and lens → indicative of PXF
    - May also see mottled pigmentation at TM (patchy/ light)
    - Packed up pigment at TM, causing the pigment to go somewhere more anterior (Sampaolesi line)
- Epidemiology: old > 70 years old, Icelandic/ nordic, systemic vascular conditions
- Risk factors: UV (put not proven), oxidative stress, i forgot the rest
- 60% cumulative risk at 5 years of progression - more aggressive and likely to convert
- Prognosis: dangers with cataracts surgery because the zonules become frosted?
- EMGT
- OHT: treatment reduces conversion to glauc
- ALT: high initial IOP brah

**westmead eye manual adiran fung → has landmark trials

PXF - more optic nerve damage and Tx more aggro
PDS - young males
POAG - if pregnant, timolol (puntatcl occlusion)
Avoid brimonadine
Know the pregnancy category
- C is better than B3 (unknown)

Drance haem ---> increased risk of NTG?

Case 4:

- 21 year old
- Irritation and itching 6 days ago?
- Both eyes, red, swollen, watery, worse in the morning
- Rubs eyes a lot
- Allergies: penicillin
- First time. Moved to Aus.
- -ve CL.
- No pain, vision not affected, no recent sickness, no trauma/ infection

Shiners -> darker, from rubbing

Clinical findings

- Entering VA: R 6/7.5 L 6/6-2
- Cornea: staining, lids for papillae or follicles, discharge, corneal sensitivity, IOP

DDx: bacterial/ allergic, viral, VKC

- Photos: upper and lower lid oedema, hyperaemic lids, on lid eversion looks like either papillae (more cobblestone like, vascular tuft) or follicles (resembles rice grains, avascular, lymphocytes)

Zaditen only BID -> >BID can worsen symptoms, more toxic side effects. Most effiacious BID no more or no less

- Palliative care: cold compress, lubricating drops (chilled)
- Patanol and Zaidten are equally effacisius. Px say patanol is more comfortable (but based on older literature).

1. Allergen avoidance
2. Combo drug
3. Low potency CS - FML bid (low penetration, low potency, low frequency, helps with surface not deeper)
    a. Can do BID (low freq)
    b. Not nsaid (stings)
    c. Wait so is the px on combo allergy + FML together
4. Refer to GP

Case 5

- 11 year old
- Bilateral itchy swollen eye
- Blurry vision
- Uncomfortable
- First ee. No known health conditions or allergies.
- Just moved to AUS.
- Watery disachagre (white), +ve photophobia.
- -ve CL

VKC -> more likely in boys by 3-4x (Australia study??) cf girls (but huge variation in diff population, dependent on humid dry hot climates)

Start with ketotifen b.i.d. + High pulse steroid dose with quick taper (because ketotifen takes 2 weeks to take effect) so once the high pulse dosage has ended, ketotifen should have kicked in

- **It usually burns out after puberty**, typically within about 4–10 years of onset, as hormonal and immune changes settle
- Estrogen receptor hyperexpression (smth.. Lily may ask)

Case 6

- 25 year old male
- 3 days **left** photophobia increasing pain
- Mild tearing
- OTC chemist drops no help

- Vision is blurry VA 6/12 in affected eye, 6/6 in non affected eye
- -ve Trauma. no previous infections.
- MHx -ve, FMHx -ve
- -ve Fxf
- -ve headaches
- -ve medications. -ve allergies.

Pressure R 16 L 11

For KPs → start off by saying deposits, then say i think it could be KPs (dont start with refined answer, esp if u wrong)

On OPTOS - inflam best seen on FAF

Lids lash - conj - cornea - AC - pupil

DDx

- Unilateral acute anterior uveitis
- Recurrent uveitis
- Chronic uveitis

How to rule out posterior? DFE/optos to check posterior involvement, OCT to check for macular swelling

Management:

- Pred forte (prednisolone acetate 1% + phenylephine 0.12%)
- Loading dose + every hour if severe
- Review next day + taper once the SUN grade drops by 2-3

If the Px shows up in the future and they have uveitis in the other eye:

- Consider review of systems
- Go to GP and refer for specific inflam marker (know the specific ones) → HLA B27 most common

We're on break now until 12:45

How is it so far??

R u guys on break yet? Or still going through cases

Aa gotcha same same

Case 7

- 55 year old nelpalasne
- Head hurts eye hurts watery
- First started 2 days ago
- Hasn't hapepned before
- No trauma
- Medications nil.
- Nausea, vomiting, painful
- Photophobia - halo around lights

VA 6/120 L 6/7.5

IOP 15 14

Goniio -> no. could induce posterio rsynechaie

Non-reactive

Mid dilated

Corneal oedema

DDx: uveitis (more miotic, constricted pupil)

Why is pilocarpine only effective if IOP < 40 - related to ischaemia of schnicter muscle

LPI - ideally within 48 hours. And also in other eye due to increase risk fo AAC

ZAP trial - reduced risk by 50% - LPI reduced the hazard of progression by **47%**, but the **absolute reduction was only about 2 percentage points** because progression was uncommon

Risk factor of angle closure: shallow AC (older px, cataracts)

Eagle study - clear lens exchange, opens up the angle.

Lens extraction vs LPI -> when to choose between the 2.

Pupillary block - primary mechanism for angle closure (aq humour cant flow anymore from posterior to anterior chamber, iris is forward, iris contact sthe lens, LPI creates alterative route for aq to flow)

Argon iridoplasty - LPI is not changing the anatomical crowding (may still get a narrow angle), if u have a plateau iris (better to do iridiolplasty or lens echange)

Signs of previous AAA

- Iris atrophy (from ischemia)
- glaucomphacin ...
- Iris vessles
- Closed angles

Case 8

32 year old male

2 days hx of sore red watery eye

Right eye

Slight photo[phobia

Slight blurred vision

-ve cl wear

First time

Mhx -ve. Did have a cold sore.

Trauma

-ve discharge

Got chrlosig from pharm - no help

-ve fxf, headache, nausea

Acute red eye

VA R 6/9 L 6/6

Pupils - normal

Facial external (vesicles)

Facial tingling → herpes, stoke

Preaurcircular lumph nodes

IOP next (after Hx) - 15 16

SLE

- Linear Dendritic lesion with oedema (corneal haze), adajacnet sataelitte lesion (Ddx herpes, fungla keratitis)
- Size (3mm = pupil size in direct illumination), 1.5mm

Ddx

- HSV epithelial keratiti s
- Acan
- Fungal
- HZO → midline respect, hutchin sign, trigeminal nerve (V1 - opthahlmic division), vesicles on top of lid and brow

1. Topical acivilor
2. Add lubricnating (esp if toxicity)

Herpetic disease -> opprtuntitis nature of herpes → MUST DILATE AND DO DFE (IF U SEE PORN OR R SYPHILIC OR NEUROTRINTIS OR CMV ETC NFLAMTORY MAKES IT AN EMERGENCY)

Debridement

Case 9

24F general chieck, feels vision slightly blurry past few months, worse in past few days, left eye affected

- Bilateral or unilateral
- Distance or near or both
- Constant or intermittent
- -ve headaches -ve nausea
- -ve itch. +ve red.
- +ve soreness, +ve photophobia
- Previous sickness. Trauma. Contact with plant or vegetatble matter

Entering VA 6/12 6/7.5

Pupils normal

IOP 16 12

Best for central MK ulcers + elderly Px: Alternate fortified dual therapy q30 mins → cephrloapsroin and gentamincin inactivate each other (form a complex) - beta lacatam from ceph will inactidatet the aminoglycoside

- Ceph break BL ring -> amino

Why no taper for AB

- Dont go under the MIC (4x day)

No gonio in AAC

If u Dont dilate herpes

If u give PGA when inflammatory

Make sure history is quick - same screening questions for acute red eye

- CL, hx of trauma, itchy, blurry

Dual therapy in MK - know MOA

When analsying photo - start with broad statement

- I am seeing smth on the cornea (instead of KP)

Can also ask if this is all the info im getting